Image Processing (1)

Shin'ichi Satoh

Purpose of this course

- Improve insight for image processing (IP)
- Find proper ways to apply IP techniques to your particular problem
- learn how to read IP papers in proper viewpoints
- Points: representation, model, and process
- Reason: human sensation and physical imaging system

What are NOT covered (or NOT addressed) by the course

- Comprehensive study in IP, from fundamentals to component techniques
- Broaden horizons in recent topics in IP
- Training to use IP techniques

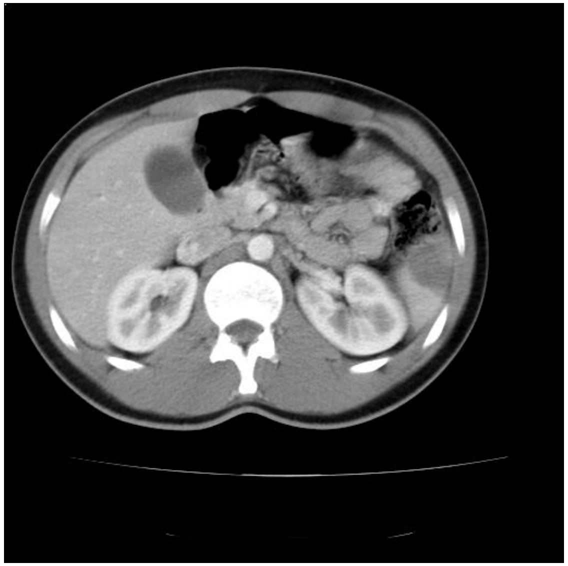
Configuration of the course

- 10 classes are planned
- time: negotiable
- first 1-2 classes: introduction and foundation
- rest: read good IP papers upon request taking volunteers (provide summary, guide discussion)
- credits: awarded by reports

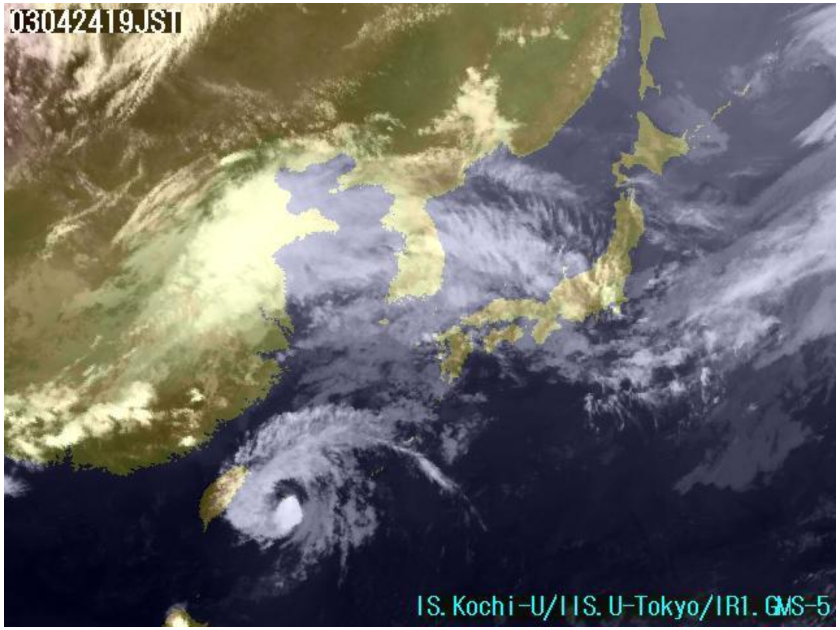
Purpose of IP

- Visualize invisible information
- Image conversion (enhancement, restoration, etc.)
- Delineate symbolic information from images (OCR, robot vision, etc.)

Visualize invisible information

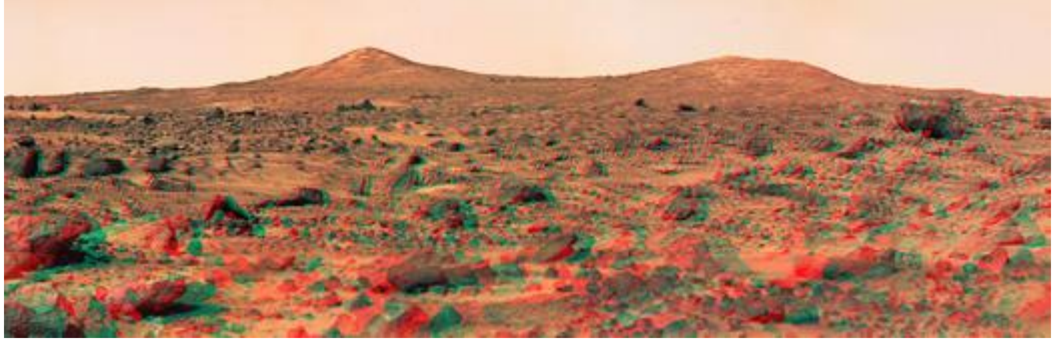


CT image



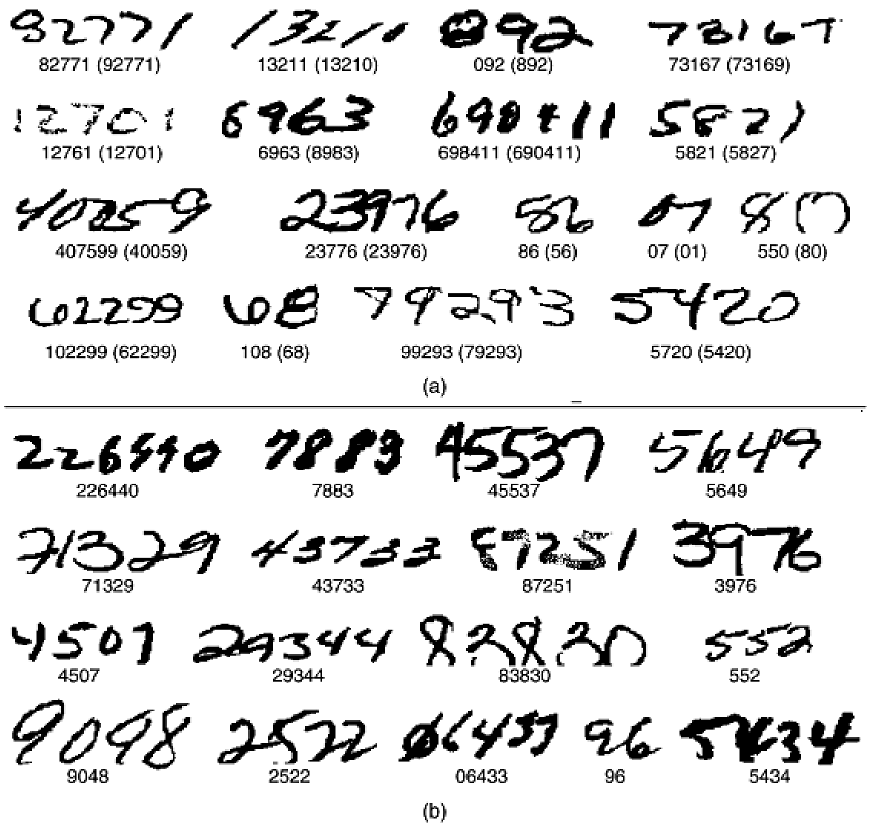
Infrared image

Image restoration and enhancement



Mars Pathfinder image (Super resolution)

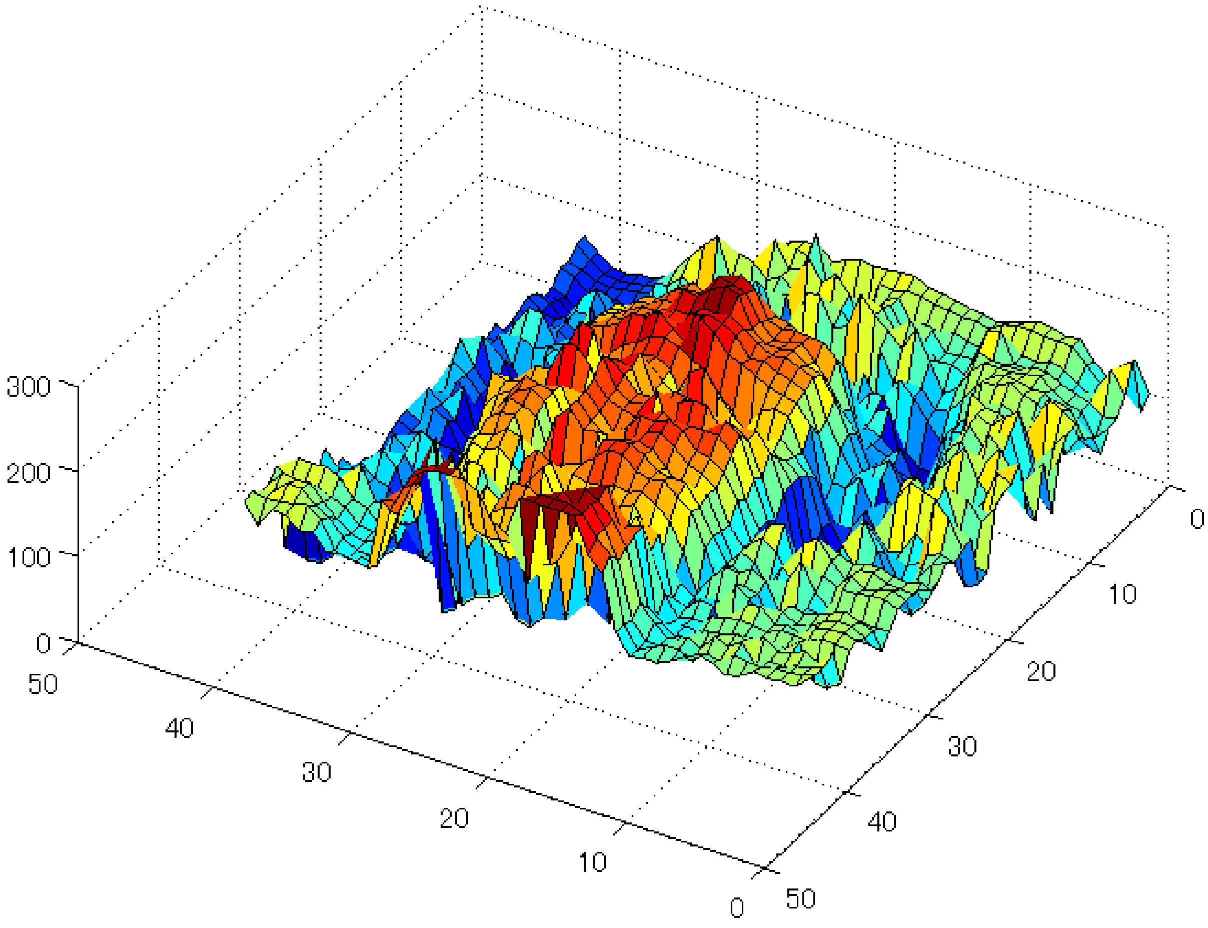
Symbolic information extraction



Why image?

- Human has too good ability in image processing.
- We can get rich information from the real world from image.
- Constitution of image is mostly influenced by two aspects:
  human's visual sensation system and
  physical system of image formulation.

Image as intensity data



Human Visual Sensation System

- Biological constitution of eyes
- Similarity to camera system
- Color sensation (discrepancies between biological sensation and physical formulation)

Recommended textbooks

- A. K. Jain, Fundamentals of Digital Image Processing, Prentice Hall, 1989.
- B. K. P. Horn, Robot Vision, MIT Press, 1986.
- A. S. Glassner, Principles of Digital Image Synthesis, Morgan Kaufmann, 1995.
- D. H. Ballard and C. M. Brown, Computer Vision, Prentice Hall, 1982.
- P. H. Lindsay and D. A. Norman, Human Information Processing: An Introduction to Psychology, Academic Press, 1977.

Recommended journals to study

- IEEE Transactions on Pattern Analysis and Machine Intelligence (TPAMI)
- IEEE Trans. on Image Processing (ToIP)
- International Journal on Computer Vision (IJCV)
- Computer Vision and Image Understanding (CVIU)
- IEEE Trans. on Multimedia (TMM)

Some papers to be read

- S. Geman and D. Geman. Stochastic relaxation, Gibbs distributions and the Bayesian restoration of images. TPAMI, 6:721--740, 1984.
- T. Gevers and A. W. M. Smeulders, PicToSeek: Combining Color and Shape Invariant Features for Image Retrieval, ToIP, 9:102-119, 2000.
- A. Srivastava, X. Liu and U. Grenander, Universal Analytical Forms for Modeling Image Probabilities, TPAMI, 24:1200-1214, 2002.
- A. Torralba and A. Oliva, Depth Estimation from Image Structure, TPAMI, 24:1226-1238, 2002.
- Y. Y. Schechnerand S. K. Nayar, Generalized Mosaicing: Wide Field of View Multispectral Imaging, TPAMI, 24:1334-1348, 2002.
- Y. Caspi and M. Irani, Spatio-Temporal Alignment of Sequences, TPAMI, 24:1409-1424, 2002.

Volunteers?